Day 10 Activity: Summarizing the `penguins`

Monday, June 1, 2026

Your Name Here

Setup

```r
library(tidyverse)
library(palmerpenguins)
```

Today's activity uses the `penguins` dataset (`palmerpenguins` package). Glimpse it before you start:

```r
glimpse(penguins)
```

```
Rows: 344
Columns: 8
$ species           <fct> Adelie, Adelie, Adelie, Adelie, Adelie, Adelie, Adel~
$ island            <fct> Torgersen, Torgersen, Torgersen, Torgersen, Torgerse~
$ bill_length_mm    <dbl> 39.1, 39.5, 40.3, NA, 36.7, 39.3, 38.9, 39.2, 34.1, ~
$ bill_depth_mm     <dbl> 18.7, 17.4, 18.0, NA, 19.3, 20.6, 17.8, 19.6, 18.1, ~
$ flipper_length_mm <int> 181, 186, 195, NA, 193, 190, 181, 195, 193, 190, 186~
$ body_mass_g       <int> 3750, 3800, 3250, NA, 3450, 3650, 3625, 4675, 3475, ~
$ sex               <fct> male, female, female, NA, female, male, female, male~
$ year              <int> 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007~
```

We will practice using **two data verbs**, `summarize()` and `group_by()`, as well as reduction functions like `n()` and `mean()`.

Part 1

For each operation below, say which family it belongs to: **reduction** (variable to one number), **transformation** (variable to one value *per row*), or **neither** (a whole-table data verb).

Table 1: Fill in the third column.

#	Operation	Family
1	The largest body mass in the dataset	[Your answer.]
2	Body mass converted from grams to kilograms	[Your answer.]
3	The number of penguins	[Your answer.]
4	Whether each penguin weighs more than 4000 g	[Your answer.]
5	The number of *distinct* islands	[Your answer.]
6	Labeling each penguin `"heavy"` or `"light"`	[Your answer.]
7	Keeping only the Gentoo penguins	[Your answer.]
8	The standard deviation of flipper length	[Your answer.]

> **Tip**
>
> The fastest test: does it return **one** number (reduction), **one per row** (transformation), or a **whole new table** (neither, that's a data verb)? Number 7 is tricky.

Part 2

2a. Write a single `summarize()` that returns the number of penguins, the mean body mass, and the mean flipper length.

```
# Your code:
```

2b. Run this, then explain in one sentence *why* the result is what it is:

```
penguins |>
  summarize(avg_mass = mean(body_mass_g))
```

```
# A tibble: 1 x 1
  avg_mass
     <dbl>
1       NA
```

[Your explanation here.]

2c. Fix 2b so it returns a real number. What one change did you make, and why does it work?

```
# Your fixed code:
```

[Your explanation here.]

Part 3

For **3a–3d**, first **write down your prediction** for how many rows the result will have and which columns it will contain. *Then* run it and check.

3a.

```
penguins |>
  group_by(species) |>
  summarize(n = n(), avg_mass = mean(body_mass_g, na.rm = TRUE))
```

Prediction (rows / columns): [Your answer.]

Actual: [Your answer.]

3b.

```
penguins |>
  group_by(species, sex) |>
  summarize(n = n(), avg_mass = mean(body_mass_g, na.rm = TRUE),
            .groups = "drop")
```

Prediction (rows / columns): [Your answer.]

Actual: [Your answer.]

Did anything about the **number of rows** surprise you? (Look closely at the `sex` column.)

[Your answer.]

3c. This one is a trap. Predict *before* you run it, then run it:

```r
penguins |>
  group_by(body_mass_g) |>
  summarize(n = n(), .groups = "drop")
```

Prediction (rows / columns): [Your answer.]

Actual: [Your answer.]

What went wrong, and what does it tell you about which *kinds* of variables you should group by?

[Your answer.]

3d. Without grouping at all:

```r
penguins |>
  summarize(n = n())
```

Prediction (rows / columns): [Your answer.]

Actual: [Your answer.]

Explain the relationship between 3a, 3b, and 3d in terms of "what does one row of the output represent?"

[Your answer.]

Part 4 — Don't trust a mean without its `n()`

Run the finest grouping we can make from these three variables:

```r
penguins |>
  group_by(species, island, sex) |>
  summarize(n = n(), avg_mass = mean(body_mass_g, na.rm = TRUE),
            .groups = "drop")
```

4a. Which group has the **fewest** penguins? Would you be comfortable quoting that group's `avg_mass` in a report? Why or why not?

[Your answer.]

4b. In one or two sentences: why is `n = n()` worth adding to *every* group summary you write, even when nobody asked for a count?

[Your answer.]

Code appendix

```
library(tidyverse)
library(palmerpenguins)
glimpse(penguins)
# Your code:

penguins |>
  summarize(avg_mass = mean(body_mass_g))
# Your fixed code:

penguins |>
  group_by(species) |>
  summarize(n = n(), avg_mass = mean(body_mass_g, na.rm = TRUE))
penguins |>
  group_by(species, sex) |>
  summarize(n = n(), avg_mass = mean(body_mass_g, na.rm = TRUE),
            .groups = "drop")
penguins |>
  group_by(body_mass_g) |>
  summarize(n = n(), .groups = "drop")
penguins |>
  summarize(n = n())
penguins |>
  group_by(species, island, sex) |>
  summarize(n = n(), avg_mass = mean(body_mass_g, na.rm = TRUE),
            .groups = "drop")
```